

Exercise J - A

1. If $y(x)$ satisfies the differential equation $y' - y \tan x = 2x \sec x$ and $y(0) = 0$, then : [JEE-2012]

यदि $y(x)$ अवकल समीकरण $y' - y \tan x = 2x \sec x$ तथा $y(0) = 0$, को सन्तुष्ट करता है तो : [JEE-2012]

(A) $y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{8\sqrt{2}}$ (B) $y'\left(\frac{\pi}{4}\right) = \frac{\pi^2}{18}$ (C) $y\left(\frac{\pi}{3}\right) = \frac{\pi^2}{9}$ (D) $y'\left(\frac{\pi}{3}\right) = \frac{4\pi}{3} + \frac{2\pi^2}{3\sqrt{3}}$

Ans. AD

Sol. $\frac{dy}{dx} - y \tan x = 2x \sec x$

I.F. $= e^{\int -\tan x \, dx} = \cos x$

$\therefore$ Equation reduces to

$$y \cdot \cos x = \int 2x \cdot \sec x \cdot \cos x \, dx$$

$$\Rightarrow y \cos x = x^2 + C$$

$$\because y(0) = 0 \Rightarrow 0 = 0 + C$$

$$\therefore y \cos x = x^2$$

$$\Rightarrow y(x) = x^2 \sec x$$

$$\therefore y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{16} \cdot \sqrt{2} = \frac{\pi^2}{8\sqrt{2}} \quad (\therefore \text{(A) is correct})$$

$$y\left(\frac{\pi}{3}\right) = \frac{\pi^2}{9} \cdot 2 = \frac{2\pi^2}{9} \quad (\therefore \text{(C) is wrong})$$

$$\text{Also } y'(x) = 2x \sec x + x^2 \sec x \tan x$$

$$\Rightarrow y'\left(\frac{\pi}{4}\right) = \frac{\pi}{2} \cdot \sqrt{2} + \frac{\pi^2 \sqrt{2}}{16} \quad (\therefore \text{(B) is wrong})$$

$$\text{and } y'\left(\frac{\pi}{3}\right) = 2 \cdot \frac{\pi}{3} \cdot 2 + \frac{\pi^2}{9} \cdot 2 \cdot \sqrt{3}$$

$$= \frac{4\pi}{3} + \frac{2\pi^2}{3\sqrt{3}} \quad (\therefore \text{(D) is correct})$$

2. A curve passes through the point $\left(1, \frac{\pi}{6}\right)$. Let the slope of the curve at each point (x, y) be $\frac{y}{x} + \sec\left(\frac{y}{x}\right)$, $x > 0$.

Then the equation of the curve is :

[JEE(Adv.)2013]

एक वक्र बिन्दु $\left(1, \frac{\pi}{6}\right)$ से गुजरता है। माना कि प्रत्येक बिन्दु (x, y) पर वक्र की प्रवणता $\frac{y}{x} + \sec\left(\frac{y}{x}\right)$, $x > 0$ है, तब वक्र का

[JEE(Adv.)2013]

(A) $\sin\left(\frac{y}{x}\right) = \log x + \frac{1}{2}$ (B) $\cos\left(\frac{y}{x}\right) = \log x + 2$ (C) $\sec\left(\frac{2y}{x}\right) = \log x + 2$ (D) $\cos\left(\frac{2y}{x}\right) = \log x + \frac{1}{2}$

Ans. A

Sol. $\frac{dy}{dx} = \frac{y}{x} + \sec \frac{y}{x}$

Let $y = vx$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + \frac{xdv}{dx} = v + \sec v$$

$$\cos v dv = \frac{dx}{x}$$

$$\sin v = \ln x + c$$

$$\sin\left(\frac{y}{x}\right) = \ln x + c$$

$$\therefore \text{passing through } \left(1, \frac{\pi}{6}\right)$$

$$\Rightarrow \sin \frac{\pi}{6} = c \Rightarrow c = \frac{1}{2}$$

$$\therefore \sin \frac{y}{x} = \ln x + \frac{1}{2}$$

3. The function $y = f(x)$ is the solution of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2 - 1} = \frac{x^4 + 2x}{\sqrt{1 - x^2}}$ in $(-1, 1)$ satisfying $f(0) = 0$.

Then: $\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x) dx$ is :

[JEE(Adv.)2014]

यदि फलन $y = f(x)$ अवकल समीकरण $\frac{dy}{dx} + \frac{xy}{x^2 - 1} = \frac{x^4 + 2x}{\sqrt{1 - x^2}}$ का अन्तराल $(-1, 1)$ में हल है जहाँ $f(0) = 0$ है तो $\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x) dx$ होगा:

[JEE(Adv.)2014]

- (A) $\frac{\pi}{3} - \frac{\sqrt{3}}{2}$ (B) $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$ (C) $\frac{\pi}{6} - \frac{\sqrt{3}}{4}$ (D) $\frac{\pi}{6} - \frac{\sqrt{3}}{2}$

Ans. B

Sol. I.F. = $e^{\int \frac{x}{x^2-1} dx} = e^{\frac{1}{2} \int \frac{2x}{x^2-1} dx} = e^{\frac{1}{2} \ln|x^2-1|} = e^{\frac{1}{2} \ln(1-x^2)} = \sqrt{1-x^2}$

$$\therefore y\sqrt{1-x^2} = \int \frac{x^4 + 2x}{\sqrt{1-x^2}} \times \sqrt{1^2 - x^2} dx + c$$

$$y\sqrt{1^2 - x^2} = \frac{x^5}{5} + x^2 + c$$

$$x=0, y=0 \Rightarrow c=0$$

$$y = \frac{\frac{x^5}{5} + x^2}{\sqrt{1-x^2}}$$

$$\therefore I = \int_0^{\frac{\sqrt{3}}{2}} \left(\frac{\frac{x^5}{5} + x^2}{\sqrt{1-x^2}} + \frac{-\frac{x^5}{5} + x^2}{\sqrt{1-x^2}} \right) dx = 2 \int_0^{\frac{\sqrt{3}}{2}} \frac{x^2}{\sqrt{1-x^2}} dx$$

$$x = \sin\theta$$

$$dx = \cos\theta d\theta$$

$$= 2 \int_0^{\frac{\pi}{3}} \frac{\sin^2\theta \cos\theta}{\cos\theta} d\theta$$

$$= \int_0^{\frac{\pi}{3}} (1 - \cos 2\theta) d\theta = \left(\theta - \frac{1}{2} \sin 2\theta \right) \Big|_0^{\frac{\pi}{3}} = \frac{\pi}{3} - \frac{1}{2} \times \frac{\sqrt{3}}{2} = \frac{\pi}{3} - \frac{\sqrt{3}}{4}$$

4. Let $y(x)$ be a solution of the differential equation $(1+e^x)y' + ye^x = 1$. If $y(0) = 2$, then which of the following statements is (are) true? **[JEE(Adv.)2015]**

- (A) $y(-4) = 0$ (B) $y(-2) = 0$
(C) $y(x)$ has a critical point in the interval $(-1, 0)$ (D) $y(x)$ has no critical point in the interval $(-1, 0)$

माना कि $y(x)$ अवकल समीकरण $(1+e^x)y' + ye^x = 1$ का हल है। यदि $y(0) = 2$ तब निम्नलिखित कथनों में से कौन सा सही है ?

[JEE(Adv.)2015]

- (A) $y(-4) = 0$ (B) $y(-2) = 0$
(C) $y(x)$ का एक क्रांतिक बिन्दु अंतराल $(-1, 0)$ में है (D) $y(x)$ का कोई भी क्रांतिक बिन्दु अंतराल $(-1, 0)$ में नहीं है

Ans. AC

Sol. $(1 + e^x) \frac{dy}{dx} + ye^x = 1$

$$\frac{dy}{dx} + \frac{e^x}{1 + e^x} y = \frac{1}{1 + e^x}$$

$$\text{I.F.} = e^{\int \frac{e^x}{1 + e^x} dx} = e^{\ln(1 + e^x)} = 1 + e^x$$

complete solution

$$y \cdot (1 + e^x) = \int 1 dx$$

$$(1 + e^x)y = x + c$$

$$x = 0, y = 2 \Rightarrow c = 4$$

$$(1 + e^x)y = x + 4$$

$$y = \frac{x + 4}{e^x + 1}$$

$$x = -4, y = 0$$

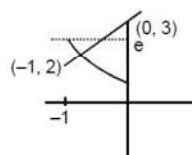
$$x = -2, y = \frac{2}{x^{-2} + 1}$$

$$\frac{dy}{dx} = \frac{(e^x + 1) \cdot 1 - (x + 4)e^x}{(e^x + 1)^2}$$

$$\frac{e^x(-x - 3) + 1}{(e^x + 1)^2}$$

$$\frac{dy}{dx} = 0 \Rightarrow x + 3 = e^{-x}$$

$$e^x = \frac{1}{x + 3}$$



5. Consider the family of all circles whose centers lie on the straight line $y = x$. If this family of circles is represented by the differential equation $Py'' + Qy' + 1 = 0$, where P, Q are functions of x, y and y' (here $y' = \frac{dy}{dx}$, $y'' = \frac{d^2y}{dx^2}$), then which of the following statements is (are) true? **[JEE(Adv.)2015]**

उन सभी वृत्त-कुल को विचार कीजिए जिनके केन्द्र सरल रेखा $y = x$ पर स्थित है। यदि इस वृत्त-कुल के सभी वृत्त, अवकल समीकरण

$$Py'' + Qy' + 1 = 0 \text{ से निरूपित होते हैं, जहाँ } P, Q \text{ इस प्रकार है कि वे } x, y \text{ और } y' \text{ के फलन है (यहाँ } y' = \frac{dy}{dx}, y'' = \frac{d^2y}{dx^2}), \text{ तब}$$

निम्नलिखित कथनों में से कौन कौन से सही है ?

[JEE(Adv.)2015]

(A) $P = y + x$

(B) $P = y - x$

(C) $P + Q = 1 - x + y + y' + (y')^2$

(D) $P - Q = x + y - y' - (y')^2$

Ans. BC

Sol. $(x - \alpha)^2 + (y - \alpha)^2 = r^2$

$$x^2 + y^2 - 2\alpha x - 2\alpha y + 2\alpha^2 - r^2 = 0$$

$$2x + 2yy' - 2\alpha - 2\alpha y' = 0 \quad \dots(i)$$

$$\Rightarrow \alpha = \frac{x + yy'}{1 + y'} \quad \dots(ii)$$

again diff. w.r.t.

$$2 + 2(y')^2 + 2yy'' - 2\alpha y'' = 0$$

$$\Rightarrow 1 + (y')^2 + yy' - \left(\frac{x + yy'}{1 + y'} \right) y'' = 0$$

$$\Rightarrow 1 + y' + (y')^2 + (y')^3 + yy'' + yy'y' - xy'' - yy'y'' = 0$$

$$\Rightarrow (y - x)y'' + (1 + y + (y')^2)y' + 1 = 0$$

$$\Rightarrow P = y - x, Q = 1 + y' + (y')^2$$

Ans. (B, C)

Note : P and Q will not be unique function

$$Py'' + Qy' + Ry' - Ry' + 1 = 0$$

$$\Rightarrow \frac{Py'}{1 - Ry''} + \frac{Qy'}{1 - Ry'} + 1 = 0 \quad \text{Hence new P \& Q can be obtained}$$

So it can be a controversial problem.

6. A solution curve of the differential equation $(x^2 + xy + 4x + 2y + 4) \frac{dy}{dx} - y^2 = 0, x > 0$, passes through the point

(1, 3). Then the solution curve :

[JEE(Adv.)2016]

(A) intersects $y = x + 2$ exactly at one point

(B) intersects $y = x + 2$ exactly at two points

(C) intersects $y = (x + 2)^2$

(D) does NOT intersect $y = (x + 3)^2$

माना कि अवकल समीकरण $(x^2 + xy + 4x + 2y + 4) \frac{dy}{dx} - y^2 = 0, x > 0$, का एक हल वक्र बिंदु (1, 3) से गुजरता है। तब वह हल

oO :

[JEE(Adv.)2016]

- (A) ल० ग 2 को ठीक एक बिंदु पर प्रतिच्छेद करता है (B) ल० ग 2 को ठीक दो बिन्दुओं पर प्रतिच्छेदित करता है
(C) ल० ग 2 को प्रतिच्छेदित करता है (D) ल० ग 3 को प्रतिच्छेदित नहीं करता है

Ans. AD

Sol. $[(x+2)^2 + y(x+2)] \frac{dy}{dx} - y^2 = 0$

Put $x+2 = t$

$$\frac{dt}{dy} = \frac{t^2 + ty}{y^2} \text{ which is homogeneous differential equation}$$

By solving it

$$\ln y = \frac{-y}{x+2} + C, C = 1 + \ln 3 \text{ since } y(1) = 3$$

$$\Rightarrow \ln\left(\frac{y}{3}\right) + \frac{y}{x+2} = 1$$

Now : Verify the alternatives.

7. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be a differentiable function such that $f'(x) = 2 - \frac{f(x)}{x}$ for all $x \in (0, \infty)$ and $f(1) \neq 1$. Then :

[JEE(Adv.)2016]

(A) $\lim_{x \rightarrow 0^+} f'\left(\frac{1}{x}\right) = 1$ (B) $\lim_{x \rightarrow 0^+} x f\left(\frac{1}{x}\right) = 2$ (C) $\lim_{x \rightarrow 0^+} x^2 f'(x) = 0$ (D) $|f(x)| \leq 2$ for all $x \in (0, 2)$

माना कि $f : (0, \infty) \rightarrow \mathbb{R}$ एक अवलकनीय फलन ऐसा है कि सभी $x \in (0, \infty)$ के लिए $f'(x) = 2 - \frac{f(x)}{x}$, और $f(1) \neq 1$ है। तब :

[JEE(Adv.)2016]

(A) $\lim_{x \rightarrow 0^+} f'\left(\frac{1}{x}\right) = 1$ (B) $\lim_{x \rightarrow 0^+} x f\left(\frac{1}{x}\right) = 2$ (C) $\lim_{x \rightarrow 0^+} x^2 f'(x) = 0$ (D) सभी $x \in (0, 2)$ के लिए $|f(x)| \leq 2$

Ans. A

Sol. $f'(x) = 2 - \frac{f(x)}{x}$ or $\frac{dy}{dx} + \frac{y}{x} = 2$

$$\text{I.F.} = e^{\int \frac{1}{x} dx} = x$$

$$\Rightarrow y \cdot x = x^2 + c \Rightarrow f(x) = x + \frac{c}{x}, c \neq 0 \text{ as } f(1) \neq 1$$

$$(A) \lim_{x \rightarrow 0^+} f'\left(\frac{1}{x}\right) = \lim_{x \rightarrow 0^+} (1 - cx^2) = 1$$

$$(B) \lim_{x \rightarrow 0^+} \left(xf\left(\frac{1}{x}\right) \right) = \lim_{x \rightarrow 0^+} (1 + cx^2) = 1$$

$$(C) \lim_{x \rightarrow 0^+} (x^2 f'(x)) = \lim_{x \rightarrow 0^+} \left(x^2 \left(1 - \frac{c}{x^2} \right) \right) = -c \neq 0$$

$$(D) \lim_{x \rightarrow 0^+} f(x) = +\infty \text{ or } -\infty, \text{ so for } c \neq 0 \text{ } f(x) \text{ is unbounded function for } x \in (0, 2)$$

8. If $y = y(x)$ satisfies the differential equation $8\sqrt{x}(\sqrt{9+\sqrt{x}})dy = \left(\sqrt{4+\sqrt{9+\sqrt{x}}}\right)^{-1} dx$, $x > 0$ and

$$y(0) = \sqrt{7}, \text{ then } y(256) =$$

[JEE(Adv.)2017]

यदि $y = y(x)$ अवकलनीय समीकरण $8\sqrt{x}(\sqrt{9+\sqrt{x}})dy = \left(\sqrt{4+\sqrt{9+\sqrt{x}}}\right)^{-1} dx$, $x > 0$ को सन्तुष्ट करता है तथा

$$y(0) = \sqrt{7} \text{ है, तब } y(256) =$$

[JEE(Adv.)2017]

$$(A) 80$$

$$(B) 9$$

$$(C) 16$$

$$(D) 3$$

Ans. D

$$\text{Sol. } \frac{dy}{dx} = \frac{\left(\sqrt{4+\sqrt{9+\sqrt{x}}}\right)^{-1}}{8\sqrt{x}\sqrt{9+\sqrt{x}}}$$

$$dy = \frac{1}{\sqrt{4+\sqrt{9+\sqrt{x}}}} \cdot \frac{1}{\sqrt{9+\sqrt{x}}} \cdot \frac{1}{8\sqrt{x}} dx$$

$$\text{Let } 4 + \sqrt{9 + \sqrt{x}} = t \Rightarrow \frac{1}{2\sqrt{9 + \sqrt{x}}} \times \frac{1}{2\sqrt{x}} dx = dt$$

$$\int dy = \int \frac{1}{\sqrt{t}} \cdot \frac{1}{2} dt$$

$$y = \sqrt{t} + c$$

$$y = \sqrt{4 + \sqrt{9 + \sqrt{x}}} + c$$

$$\text{at } x = 0, y = \sqrt{7}$$

$$\Rightarrow \sqrt{7} = \sqrt{7} + c \Rightarrow c = 0$$

$$\text{at } x = 256 \Rightarrow y = \sqrt{4 + \sqrt{9 + \sqrt{256}}} = 3$$

9. If $f: \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function such that $f'(x) > 2f(x)$ for all $x \in \mathbb{R}$, and $f(0) = 1$, then :

[JEE(Adv.)2017]

(A) $f(x)$ is increasing in $(0, \infty)$

(B) $f'(x) < e^{2x}$ in $(0, \infty)$

(C) $f(x) > e^{2x}$ in $(0, \infty)$

(D) $f(x)$ is decreasing in $(0, \infty)$

यदि $f: \mathbb{R} \rightarrow \mathbb{R}$ इस प्रकार का अवकलनीय फलन है, कि सभी $x \in \mathbb{R}$ के लिये $f'(x) > 2f(x)$ एवं $f(0) = 1$ है, तब :

[JEE(Adv.)2017]

(A) $(0, \infty)$ में $f(x)$ वर्धमान है

(B) $(0, \infty)$ में $f'(x) < e^{2x}$

(C) $(0, \infty)$ में $f(x) > e^{2x}$

(D) $(0, \infty)$ में $f(x)$ ह्रासमान है

Ans. AC

Sol. $f'(x) - 2f(x) > 0$

$$\Rightarrow \frac{d}{dx} (f(x) \cdot e^{-2x}) > 0 \quad \Rightarrow \quad g(x) = f(x) \cdot e^{-2x} \text{ is an increasing function}$$

$$\text{for } x > 0, \quad g(x) > g(0)$$

$$\Rightarrow f(x) \cdot e^{-2x} > 1 \quad \Rightarrow \quad f(x) > e^{2x}$$

$$\text{Now } f'(x) > 2f(x) > 2 \cdot e^{2x}$$

10. Let $f: \mathbb{P} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be two non-constant differentiable functions. If $f'(x) = (e^{(f(x)-g(x))})g'(x)$ for all $x \in \mathbb{R}$ and $f(1) = g(2) = 1$, then which of the following statement(s) is (are) TRUE? [JEE(Adv.)2018]

(A) $f(2) < 1 - \log_e 2$ (B) $f(2) > 1 - \log_e 2$ (C) $g(1) > 1 - \log_e 2$ (D) $g(1) < 1 - \log_e 2$

माना कि $f: \mathbb{R} \rightarrow \mathbb{R}$ और $g: \mathbb{R} \rightarrow \mathbb{R}$ दो चर और अवकलनीय फलन हैं। यदि $f'(x) = (e^{(f(x)-g(x))})g'(x)$ सभी $x \in \mathbb{R}$ के लिये और $f(1) = g(2) = 1$, तब निम्नलिखित में से कौनसा (से) कथन सत्य है (हैं)? [JEE(Adv.)2018]

(A) $f(2) < 1 - \log_e 2$ (B) $f(2) > 1 - \log_e 2$ (C) $g(1) > 1 - \log_e 2$ (D) $g(1) < 1 - \log_e 2$

Ans. BC

Sol. $f'(x) = e^{f(x)-g(x)} g'(x) : f(1) = g(2) = 1$

$$e^{-f(x)} = e^{-g(x)} + C$$

$$e^{-f(x)} \cdot f'(x) = e^{-g(x)} \cdot g'(x)$$

$$\int d(e^{-f(x)}) = \int d(e^{-g(x)})$$

$$e^{-f(x)} = e^{-g(x)} + c$$

$$x=1 \quad \frac{1}{e} = e^{-g(1)} + c$$

$$x=2 \quad e^{-f(2)} = \frac{1}{e} + c$$

$$\therefore g(1) > 1 - \ln 2$$

$$e^{-f(2)} = 2e^{-1} - e^{-g(1)}$$

$$e^{-f(2)} = 2e^{-1} - e^{-g(1)}$$

$$f(2) > 1 - \ln 2$$

$$e^{-1} - e^{-f(2)} = e^{-g(1)} - e^{-1} \Rightarrow e^{-g(1)} + e^{-f(2)} = 2e^{-1}$$

$$e^{-g(1)} < 2e^{-1}$$

$$-g(1) < \ln 2 - 1$$

11. Let $f: (0, \pi) \rightarrow \mathbb{R}$ be a twice differentiable function such that $\lim_{t \rightarrow x} \frac{f(x)\sin t - f(t)\sin x}{t - x} = \sin^2 x$ for all $x \in (0, \pi)$. If

$$f\left(\frac{\pi}{6}\right) = -\frac{\pi}{12}, \text{ then which of the following statement(s) is (are) TRUE?}$$

[JEE(Adv.)2018]

$$(A) f\left(\frac{\pi}{4}\right) = \frac{\pi}{4\sqrt{2}}$$

$$(B) f(x) < \frac{x^4}{6} - x^2 \text{ for all } x \in (0, \pi)$$

$$(C) \text{ There exists } \alpha \in (0, \pi) \text{ such that } f(\alpha) = 0 \quad (D) f''\left(\frac{\pi}{2}\right) + f\left(\frac{\pi}{2}\right) = 0$$

माना कि $f: (0, \pi) \rightarrow \mathbb{R}$ एक ऐसा द्विअवकलनीय फलन है कि $\lim_{t \rightarrow x} \frac{f(x)\sin t - f(t)\sin x}{t - x} = \sin^2 x$ सभी $x \in (0, \pi)$ के लिये। यदि

$$f\left(\frac{\pi}{6}\right) = -\frac{\pi}{12}, \text{ तब निम्नलिखित में से कौन सा (से) कथन सत्य है (हैं)?}$$

[JEE(Adv.)2018]

$$(A) f\left(\frac{\pi}{4}\right) = \frac{\pi}{4\sqrt{2}}$$

$$(B) f(x) < \frac{x^4}{6} - x^2 \text{ सभी } x \in (0, \pi) \text{ के लिये}$$

$$(C) \text{ एक ऐसे } \alpha \in (0, \pi) \text{ का अस्तित्व है जिसके लिये } f(\alpha) = 0 \quad (D) f''\left(\frac{\pi}{2}\right) + f\left(\frac{\pi}{2}\right) = 0$$

Ans. BCD

Sol. $\lim_{t \rightarrow x} \frac{f(x)\sin t - f(t)\sin x}{t - x} = \sin^2 x$

$$\frac{f(x)\cos x - f'(x)\sin x}{\sin^2 x} = 1$$

$$-d\left(\frac{f(x)}{\sin x}\right) = 1$$

$$\frac{f(x)}{\sin x} = -x + c \quad \because f\left(\frac{\pi}{6}\right) = -\frac{\pi}{12} \Rightarrow c = 0 \Rightarrow f(x) = -x \sin x$$

$$(D) f(x) + f''(x) = -2\cos x$$

$$f\left(\frac{\pi}{2}\right) + f''\left(\frac{\pi}{2}\right) = 0$$

$$(A) f\left(\frac{\pi}{4}\right) = \left(-\frac{\pi}{4\sqrt{2}}\right)$$

$$(C) f(x) \text{ is continuous and differentiable and } f(0) = f(x) = 0$$

Using LMVT $f'(c) = 0$ for some $x \in (0, \pi)$

$$(B) g(x) = -x \sin x + x^2 - \frac{x^4}{6}$$

$$g'(x) = f'(x) + 2x - \frac{2x^3}{3}$$

$$g''(x) = f''(x) + 2x - 2x^2$$

$$g'''(x) = 3 \sin x + x \cos x - 4x = 3(\sin x - x) + x(\cos x - 1)$$

$$\Rightarrow g'''(x) < 0 \Rightarrow g''(x) \text{ is decreasing}$$

$$\text{for } x > 0 \quad g''(x) < g''(0) \Rightarrow g''(x) < 0$$

hence $g'(x)$ is decreasing

$$\text{for } x > 0 \quad g'(x) < g'(0) \Rightarrow g'(x) < 0$$

hence $g(x) < 0$

$$\text{for } x > 0 \quad g(x) < g(0) \Rightarrow g(x) < 0$$

$$\text{Hence } f(x) < \frac{x^4}{6} - x^2 \forall x \in (0, \pi)$$

12. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = 0$. If $y = f(x)$ satisfies the differential equation

$$\frac{dy}{dx} = (2 + 5y)(5y - 2), \text{ then the value of } \lim_{x \rightarrow -\infty} f(x) \text{ is } \underline{\hspace{2cm}}.$$

[JEE(Adv.)2018]

माना कि $f: \mathbb{R} \rightarrow \mathbb{R}$ एक ऐसा अवकलनीय फलन है जिसके लिये $f(0) = 0$. यदि $y = f(x)$, अवकल समीकरण $\frac{dy}{dx} = (2 + 5y)(5y - 2)$

को सन्तुष्ट करता है, तब $\lim_{x \rightarrow -\infty} f(x)$ का मान है।

[JEE(Adv.)2018]

Ans. 0.4

Sol. $\frac{dy}{dx} = (5y+2)(5y-2)$

$$\frac{1}{25} \int \frac{dy}{\left(y + \frac{2}{5}\right)\left(y - \frac{2}{5}\right)} = \int dx$$

$$\frac{1}{25} \cdot \frac{5}{4} \ln \left| \frac{y - \frac{2}{5}}{y + \frac{2}{5}} \right| = x + c$$

$$\frac{1}{20} \ln \left| \frac{5y-2}{5y+2} \right| = x + c$$

at $x = 0, y = 0 \Rightarrow c = 0$

Hence $\frac{2-5y}{2+5y} = e^{20x}$

$$\frac{2-5y}{2+5y} = e^{20x}, \quad \lim_{x \rightarrow -\infty} e^{20x} = 0 \quad \Rightarrow \quad \lim_{x \rightarrow -\infty} y = \frac{2}{5} = 0.4$$

13. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = 1$ and satisfying the equation $f(x+y) = f(x)f(y) + f'(x)f(y)$ for all $x, y \in \mathbb{R}$. Then, the value of $\log_e(f(4))$ is _____. **[JEE(Adv.)2018]**

माना कि $f: \mathbb{R} \rightarrow \mathbb{R}$ एक ऐसा अवकलनीय फलन है जिसके लिये $f(0) = 1$, और जो सभी $x, y \in \mathbb{R}$ के लिये समीकरण $f(x+y) = f(x)f(y) + f'(x)f(y)$ को सन्तुष्ट करता है। तब $\log_e(f(4))$ का मान है। **[JEE(Adv.)2018]**

Ans. 2

Sol. $f(x+y) = f(x) \cdot f'(y) + f'(x) \cdot f(y)$
substituting $x = y = 0$, we get

$$f(0) = 2f'(0) \quad \Rightarrow \quad f'(0) = \frac{1}{2}$$

Now substituting $y = 0$

$$f(x) = f(x) \cdot f'(0) + f'(x) \cdot f(0)$$

$$\Rightarrow f'(x) = \frac{f(x)}{2}$$

$$\Rightarrow f'(x) = \lambda e^{\frac{x}{2}} \quad \Rightarrow \quad f(x) = e^{\frac{x}{2}} \quad (\text{as } f(0) = 1)$$

$$\text{Now } \ln(f(x)) = \frac{x}{2} \Rightarrow \ln(f(4)) = 2$$

14. Let Γ denote a curve $y = y(x)$ which is in the first quadrant and let the point $(1, 0)$ on it. Let the tangent to Γ at a point P intersect the y -axis at Y_p . If PY_p has length 1 for each point P on Γ , then which of the following options is/are correct : [JEE(Adv.)2019]

माना कि Γ एक वक्र $y = y(x)$ है जो प्रथम चतुर्थांश (first quadrant) में है और माना कि बिन्दु $(1, 0)$ उस पर स्थित है। माना कि Γ के बिन्दु P पर खींची गयी स्पर्श रेखा (tangent) y -अक्ष को Y_p पर प्रतिच्छेद (intersect) करती है, यदि Γ के प्रत्येक बिन्दु P के लिए PY_p की लम्बाई 1 है। तब निम्न में से कौन सा (से) कथन सही है : [JEE(Adv.)2019]

(A) $xy' + \sqrt{1-x^2} = 0$

(B) $y = -\log_e \left(\frac{1 + \sqrt{1-x^2}}{x} \right) + \sqrt{1-x^2}$

(C) $y = \log_e \left(\frac{1 + \sqrt{1-x^2}}{x} \right) - \sqrt{1-x^2}$

(D) $xy' - \sqrt{1-x^2} = 0$

Ans. AC

Sol. Equation of tangent at P

$$Y - y = m(X - x)$$

$$Y_p(0, y - mx)$$

$$PY_p = 1$$

$$\sqrt{x^2 + m^2 x^2} = 1$$

$$x^2 + m^2 x^2 = 1$$

$$m^2 = \frac{1-x^2}{x^2}$$

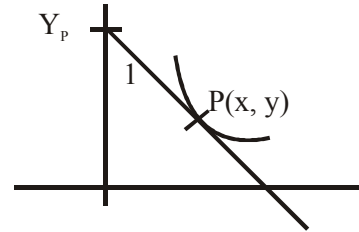
$$m = \pm \sqrt{\frac{1-x^2}{x^2}}$$

$$\frac{dy}{dx} = \frac{\sqrt{1-x^2}}{x}$$

$$\int dy = \int \frac{\sqrt{1-x^2}}{x} dx$$

$$\text{Put } x = \sin \theta$$

$$y = \int \frac{\cos^2 \theta}{\sin \theta} d\theta$$



$$\frac{dy}{dx} = -\frac{\sqrt{1-x^2}}{x}$$

$$\int dy = -\int \frac{\sqrt{1-x^2}}{x} dx$$

$$\text{Put } x = \sin \theta$$

$$y = \ln \left(\frac{1 + \sqrt{1-x^2}}{x} \right) - \sqrt{1-x^2} + c$$

$$\Rightarrow c = 0$$

$$\begin{aligned}
&= \int \frac{1 - \sin^2 \theta}{\sin \theta} d\theta \\
&= \int (\operatorname{cosec} \theta - \sin \theta) d\theta \\
&= -\ln(\operatorname{cosec} \theta + \cot \theta) + \cos \theta + c \\
&= -\ln\left(\frac{1 + \sqrt{1-x^2}}{x}\right) + \sqrt{1-x^2} + c \\
&\Rightarrow c = 0
\end{aligned}$$

(Rejected because curve is lying in IV Quadrant)

15. Let b be a nonzero real number. Suppose $f: \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function such that $f(0) = 1$. If the derivative f' of f satisfies the equation $f'(x) = \frac{f(x)}{b^2 + x^2}$ for all $x \in \mathbb{R}$, then which of the following statements is/are TRUE?

[JEE(Adv.)2020]

- (A) If $b > 0$, then f is an increasing function (B) If $b < 0$, then f is a decreasing function
 (C) $f(x)f(-x) = 1$ for all $x \in \mathbb{R}$ (D) $f(x) - f(-x) = 0$ for all $x \in \mathbb{R}$

माना कि b एक शून्येतर वास्तविक संख्या है। मान लीजिये कि $f: \mathbb{R} \rightarrow \mathbb{R}$ एक ऐसा अवकलनीय फलन है जिसके लिये $f(0) = 1$ है।

यदि f का अवकलज f' समीकरण $f'(x) = \frac{f(x)}{b^2 + x^2}$ को सभी $x \in \mathbb{R}$ के लिये संतुष्ट करता है, तब निम्न में से कौनसा (से) कथन सही (हैं)?

[JEE(Adv.)2020]

- (A) यदि $b > 0$ है, तब f एक वर्धमान फलन है (B) यदि $b < 0$ है, तब f एक ह्रासमान फलन है
 (C) सभी $x \in \mathbb{R}$ के लिए $f(x)f(-x) = 1$ है (D) सभी $x \in \mathbb{R}$ के लिए $f(x) - f(-x) = 0$ है

Ans. AC

Sol. f is non zero

$$\int \frac{f'(x)}{f(x)} dx = \int \frac{1}{x^2 + b^2} dx$$

$$\ln(f(x)) = \frac{1}{b} \tan^{-1} \frac{x}{b} + c$$

$$0 = 0 + c \quad \{\because f(0) = 1\}$$

$$f(x) = e^{\frac{1}{b} \tan^{-1} \frac{x}{b}}$$

$$f(-x) = e^{\frac{-1}{b} \tan^{-1} \frac{x}{b}}$$

(A) $f(x)$ is positive function for all $x \in \mathbb{R}$

$$f'(x) = \frac{f(x)}{x^2 + b^2}$$

$f'(x)$ is also positive function

so $f(x)$ is increasing function

$$(d) f(x) - f(-x) = e^{\frac{1}{b} \tan^{-1} \frac{x}{b}} - e^{\frac{-1}{b} \tan^{-1} \frac{x}{b}} \neq 0$$

for all $x \in \mathbb{R}$

$$(c) f(x) f(-x) = 1$$

$$e^0 = 1$$

$$1 = 1$$

16. For any real numbers α and β let $y_{\alpha, \beta}$, $x \in \mathbb{R}$, be the solution of the differential equation $\frac{dy}{dx} + \alpha y = x e^{\beta x}$, $y(1) = 1$. Let $S = \{y_{\alpha, \beta}(x) : \alpha, \beta \in \mathbb{R}\}$. Then which of the following functions belong(s) to the set S ?

[JEE(Adv.)2021]

किन्हीं वास्तविक संख्याओं α व β के लिए, माना $y_{\alpha, \beta}$, $x \in \mathbb{R}$ अवकल समीकरण $\frac{dy}{dx} + \alpha y = x e^{\beta x}$, $y(1) = 1$ का हल है। माना

$S = \{y_{\alpha, \beta}(x) : \alpha, \beta \in \mathbb{R}\} \subset \mathbb{R}$, तब निम्न में से कौनसा/कौनसे फलन समुच्चय S से सम्बन्धित है/हैं? [JEE(Adv.)2021]

$$(A) f(x) = \frac{x^2}{2} e^{-x} + \left(e - \frac{1}{2}\right) e^{-x}$$

$$(B) f(x) = -\frac{x^2}{2} e^{-x} + \left(e + \frac{1}{2}\right) e^{-x}$$

$$(C) f(x) = \frac{e^x}{2} \left(x - \frac{1}{2}\right) + \left(e - \frac{e^2}{4}\right) e^{-x}$$

$$(D) f(x) = \frac{e^x}{2} \left(\frac{1}{2} - x\right) + \left(e + \frac{e^2}{4}\right) e^{-x}$$

Ans. AC

Sol. $\frac{dy}{dx} + \alpha y = x e^{\beta x}$

$$I.F. = e^{\int \alpha dx} = e^{\alpha x}$$

$$y \cdot e^{\alpha x} = \int e^{\alpha x} \cdot (x e^{\beta x}) dx + C$$

$$y \cdot e^{\alpha x} = \int x \cdot e^{(\alpha + \beta)x} dx + C$$

$$y \cdot e^{\alpha x} = \frac{x \cdot e^{(\alpha + \beta)x}}{(\alpha + \beta)} - \int \frac{e^{(\alpha + \beta)x}}{\alpha + \beta} dx + C$$

$$y.e^{\alpha x} = e^{(\alpha+\beta)x} \left[\frac{x}{\alpha+\beta} - \frac{1}{(\alpha+\beta)^2} \right] + C$$

$$y(1) = 1$$

$$\Rightarrow e^{\alpha} = e^{\alpha+\beta} \left[\frac{1}{\alpha+\beta} - \frac{1}{(\alpha+\beta)^2} \right] + C$$

$$C = \left(1 - e^{\beta} \left(\frac{1}{\alpha+\beta} - \frac{1}{(\alpha+\beta)^2} \right) \right) e^{\alpha}$$

$$y.e^{\alpha x} = e^{(\alpha+\beta)x} \left[\frac{x}{\alpha+\beta} - \frac{1}{(\alpha+\beta)^2} \right] + e^{\alpha} \left(1 - e^{\beta} \left(\frac{1}{\alpha+\beta} - \frac{1}{(\alpha+\beta)^2} \right) \right)$$

$$y = e^{\beta x} \left(\frac{x}{\alpha+\beta} - \frac{1}{(\alpha+\beta)^2} \right) + \left(e^{\alpha} - e^{\alpha+\beta} \left(\frac{1}{\alpha+\beta} - \frac{1}{(\alpha+\beta)^2} \right) \right) e^{-\alpha x}$$

$$\text{for } \alpha = \beta = 1$$

$$y = \frac{e^x}{2} \left(x - \frac{1}{2} \right) + \left(e - \frac{e^2}{4} \right) e^{-x}$$

$$\text{for } \alpha = 1, \beta = -1$$

$$\frac{dy}{dx} + y = x.e^{-x}$$

$$\text{IF } e^{\int dx} = e^x$$

$$y.e^x = \int x dx = \frac{x^2}{2} + C$$

$$y(1) = 1$$

$$| \vec{OB} \times \vec{OC} | = \frac{9}{2}$$

$$y.e^x = \frac{x^2}{2} + e - \frac{1}{2}$$

$$y = \frac{x^2}{2}.e^{-x} + \left(e - \frac{1}{2} \right) e^{-x}$$

$$\text{As (A, C)}$$

17. If $y(x)$ is the solution of the differential equation $xdy - (y^2 - 4y)dx = 0$ for $x > 0$, $y(1) = 2$, and the slope of the curve $y = y(x)$ is never zero, then the value of $10y(\sqrt{2})$ is _____. [JEE(Adv.)2022]

यदि $x > 0$ के लिए, $y(x)$ अवकल समीकरण $xdy - (y^2 - 4y)dx = 0$ for $x > 0$, $y(1) = 2$, का हल है एवं वक्र $y = y(x)$ का ढाल कभी भी शून्य नहीं होता है, तब $10y(\sqrt{2})$ का मान _____ है। [JEE(Adv.)2022]

Ans. 8

Sol. $xdy - (y^2 - 4y)dx = 0, x > 0$

$$\int \frac{dy}{y^2 - 4y} = \int \frac{dx}{x}$$

$$\int \left(\frac{1}{y-4} - \frac{1}{y} \right) dy = 4 \int \frac{dx}{x}$$

$$\log_e |y-4| - \log_e |y| = 4 \log_e x + \log_e c$$

$$\frac{|y-4|}{|y|} = cx^4 \xrightarrow{(1,2)} c = 1$$

$$|y-4| = |y| x^4$$

C-1

and C-2

$$y-4 = yx^4$$

$$y-4 = -yx^4$$

$$y = \frac{4}{1-x^4}$$

$$y = \frac{4}{1+x^4}$$

$$y(1) = \text{ND (rejected)}$$

$$y(1) = 2$$

$$y(\sqrt{2}) = \frac{4}{5} \Rightarrow 10y(\sqrt{2}) = 8$$

18. For $x \in \mathbb{R}$, let the function $y(x)$ be the solution of the differential equation $\frac{dy}{dx} + 12y = \cos\left(\frac{\pi}{12}x\right)$, $y(0) = 0$. Then,

which of the following statements is/are TRUE?

[JEE(Adv.)2022]

(A) $y(x)$ is an increasing function

(B) $y(x)$ is decreasing function

(C) There exists a real number β such that the line $y = \beta$ intersects the curve $y = y(x)$ at infinitely many points

(D) $y(x)$ is a periodic function

माना कि $x \in \mathbb{R}$ के लिए, फलन $y(x)$ अवकल समीकरण $\frac{dy}{dx} + 12y = \cos\left(\frac{\pi}{12}x\right)$, $y(0) = 0$ का हल है। तब निम्न कथनों में से

कौनसा सत्य है?

[JEE(Adv.)2022]

- (A) ल गद्धएक वर्धमान फलन है
 (B) ल गद्धएक ह्यसमान फलन है
 (C) एक ऐसी वास्तविक संख्या β है जिसके लिए रेखा $y = \beta$, वक्र $y = y(x)$ को अनंत बिंदुओं पर प्रतिच्छेदित करती है
 (D) ल गद्धएक आवर्ती फलन है

Ans. C

Sol. $\frac{dy}{dx} + 12y = \cos\left(\frac{\pi x}{12}\right)$

$$\text{I.F.} = e^{12x} \Rightarrow y \cdot e^{12x} = \int e^{12x} \cdot \cos\left(\frac{\pi x}{12}\right) dx + C$$

$$\Rightarrow y \cdot e^{12x} = \frac{e^{12x}}{12^2 + \left(\frac{\pi}{12}\right)^2} \left[12 \cos \frac{\pi x}{12} + \frac{\pi}{12} \sin \left(\frac{\pi x}{12} \right) \right]$$

$$\because y(0) = 0 \Rightarrow C = -\frac{12}{12^2 + \left(\frac{\pi}{12}\right)^2}$$

$$\text{So } y = \frac{1}{\lambda} \underbrace{\left[12 \cos \left(\frac{\pi x}{12} \right) + \frac{\pi}{12} \sin \left(\frac{\pi x}{12} \right) \right]}_{f_1(x)} - 12e^{-12x}$$

$$\frac{dy}{dx} = \frac{1}{\lambda} \underbrace{\left[-\pi \sin \left(\frac{\pi x}{12} \right) + \frac{\pi^2}{12^2} \cos \left(\frac{\pi x}{12} \right) \right]}_{f_2(x)} + 12e^{-12x}$$

When x is large then $12e^{-12x}$ tends to zero.

$$\text{But } f_2(x) \text{ varies in } \left[-\sqrt{\pi^2 + \left(\frac{\pi}{12}\right)^4}, \sqrt{\pi^2 + \left(\frac{\pi}{12}\right)^4} \right]$$

Hence $\frac{dy}{dx}$ is changing its sign.

So $y(x)$ is non monotonic for all real number.

Also when x is very large then again $-12e^{-12x}$ is almost zero but $f_1(x)$ is periodic, so there exist some β for which $y = \beta$ intersect $y = y(x)$ at infinitely many points.

19. For $x \in \mathbb{R}$, let $y(x)$ be a solution of the differential equation $(x^2 - 5) \frac{dy}{dx} - 2xy = -2x(x^2 - 5)^2$ such that $y(2) = 7$. Then the maximum value of the function $y(x)$ is _____. [JEE(Adv.)2023]

माना कि $y(x)$, $x \in \mathbb{R}$ निम्नलिखित अवकल समीकरण (differential equation) $(x^2 - 5) \frac{dy}{dx} - 2xy = -2x(x^2 - 5)^2$ का एक
 , $\frac{1}{2}$ kg $\frac{1}{2}$ d sfy, $y(2) = 7$ है। तब फलन $y(x)$ का महत्तम मान (maximum value) है _____ [JEE(Adv.)2023]

Ans. 16

Sol. $\frac{dy}{dx} - \frac{2x}{(x^2 - 5)} y = -2x(x^2 - 5)$

$$\text{If } e^{-\int \frac{2x}{x^2-5} dx} = e^{-\ln|x^2-5|} = \frac{1}{|x^2-5|}$$

Solution of DE

$$y \cdot \frac{1}{|x^2-5|} = \int \frac{-2x(x^2-5)}{|x^2-5|} dx + C$$

$$\Rightarrow y \cdot \frac{1}{|x^2-5|} = -x^2 \left(\frac{x^2-5}{|x^2-5|} \right) + C$$

$$y(2) = 7$$

$$\Rightarrow 7 = 4 + C \quad \Rightarrow \boxed{C = 3}$$

$$\text{So } y = -x^2(x^2 - 5) + 3|x^2 - 5|$$

$$\text{if } -\sqrt{5} < x < \sqrt{5}$$

$$y = (5 - x^2)(x^2 + 3)$$

$$= -x^4 + 2x^2 + 15$$

$$= -[x^4 - 2x^2 - 15]$$

$$= -[(x^2 - 1)^2 - 16]$$

$$= 16 - (x^2 - 1)^2$$

$$\text{So max at } x = \pm 1$$

$$\text{and } \boxed{y_{\max} = 16}$$

$$\text{If } |x| > \sqrt{5}$$

$$y = (x^2 - 5)(-x^2 + 3)$$

$$= -x^4 + 8x^2 - 15$$

$$= -[x^4 - 8x^2 + 15]$$

$$= -[(x^2 - 4)^2 - 1]$$

$$= 1 - (x^2 - 4)^2$$

$$\boxed{\text{Ans} = 16}$$

20. Let $f(x)$ be a continuously differentiable function on the interval $(0, \infty)$ such that $f(1) = 2$ and $\lim_{t \rightarrow x} \frac{t^{10}f(x) - x^{10}f(t)}{t^9 - x^9} = 1$
 for each $x > 0$. Then, for all $x > 0$, $f(x)$ is equal to : [JEE(Adv.)2024]

माना कि $f(x)$, अंतराल (interval) $(0, \infty)$ में सांतत्य रूप से अवकलनीय (continuously differentiable) एक इस प्रकार का फलन

(function) है कि $f(1) = 2$, तथा प्रत्येक $x > 0$ के लिए $\lim_{t \rightarrow x} \frac{t^{10}f(x) - x^{10}f(t)}{t^9 - x^9} = 1$ है। तब सभी $x > 0$ के लिए $f(x)$ बराबर है :

[JEE(Adv.)2024]

(A) $\frac{31}{11x} - \frac{9}{11}x^{10}$ (B) $\frac{9}{11x} + \frac{13}{11}x^{10}$ (C) $\frac{-9}{11x} + \frac{31}{11}x^{10}$ (D) $\frac{13}{11x} + \frac{9}{11}x^{10}$

Ans. B

Sol. $\lim_{t \rightarrow x} \frac{t^{10}f(x) - x^{10}f(t)}{t^9 - x^9} = 1 \quad (x > 0)$

Apply L-H rule

$$\lim_{t \rightarrow x} \frac{10t^9f(x) - x^{10}f'(t)}{9t^8} = 1$$

$$\Rightarrow 10x f(x) - x^2 f'(x) = 9$$

$$x^2 \frac{dy}{dx} - 10xy = -9$$

$$\frac{dy}{dx} - \frac{10}{x}y = -\frac{9}{x^2}$$

$$\text{I.F.} = e^{\int -\frac{10}{x} dx} = \frac{1}{x^{10}}$$

$$y \times \frac{1}{x^{10}} = \frac{-9}{-11(x^{11})} + C$$

$$\frac{y}{x^{10}} = \frac{9}{11x^{11}} + C$$

$$f(1) = 2 \text{ put } x = 1, y = 2$$

$$2 = \frac{9}{11} + C$$

$$C = \frac{13}{11}$$

$$\frac{y}{x^{10}} = \frac{9}{11x^{11}} + \frac{13}{11}$$

$$y = \frac{9}{11x} + \frac{13}{11}x^{10}$$

CHALLENGER EXERCISES

Exercise C - I

SINGLE CORRECT TYPE QUESTIONS

1. A function $y = f(x)$ satisfies the differential equation $f(x) \cdot \sin 2x - \cos x + (1 + \sin^2 x) f'(x) = 0$ with initial condition $y(0) = 0$. The value of $f\left(\frac{\pi}{6}\right)$ is equal to :

फलन $y = f(x)$, अवकल समीकरण $f(x) \cdot \sin 2x - \cos x + (1 + \sin^2 x) f'(x) = 0$ को सन्तुष्ट करता है। जबकि प्रारम्भिक अवस्था

$y(0) = 0$ । तो $f\left(\frac{\pi}{6}\right)$ का मान होगा :

- (A) $1/5$ (B) $3/5$ (C) $4/5$ (D) $2/5$

Ans. D

Sol. $y \sin 2x - \cos x + (1 + \sin^2 x) \frac{dy}{dx} = 0$ where $y = f(x)$

$$\frac{dy}{dx} + \left(\frac{\sin 2x}{1 + \sin^2 x} \right) y = \frac{\cos x}{1 + \sin^2 x}$$

$$\text{I.F.} = e^{\int \frac{\sin 2x}{1 + \sin^2 x} dx} = e^{\int \frac{dt}{t}} = e^{\ln(1 + \sin^2 x)} = 1 + \sin^2 x \quad (\text{by putting } 1 + \sin^2 x = t)$$

$$y(1 + \sin^2 x) = \int \cos x dx$$

$$y(1 + \sin^2 x) = \sin x + C ; \quad (y(0) = 0) \Rightarrow C = 0$$

$$\text{hence, } y = \frac{\sin x}{1 + \sin^2 x}$$

$$y\left(\frac{\pi}{6}\right) = \frac{2}{5}$$

2. Water is drained from a vertical cylindrical tank by opening a valve at the base of the tank. It is known that the rate at which the water level drops is proportional to the square root of water depth y , where the constant of proportionality $k > 0$ depends on the acceleration due to gravity and the geometry of the hole. If t is measured in minutes and $k = \frac{1}{15}$ then the time to drain the tank if the water is 4 meter deep to start with is :

उर्ध्वाधर बेलनाकार टंकी से, उसके आधार पर स्थित छेद (hole) से, पानी बाहर निकल रहा है। यह ज्ञात है कि टंकी में पानी का तल गिरने की दर किसी भी समय पर पानी के गहराई y के वर्ग मूल के समानुपाती है। जहाँ समानुपाती स्थिरांक $k > 0$ गुरुत्वीय त्वरण का तथा छेद

(hole) के आकार पर निर्भर करता है। यदि $k = \frac{1}{15}$ तथा t मिनट में हो तो टंकी के खाली होने में लगा समय कितना होगा जहाँ शुरू में

पानी की गहराई 4 मीटर का है :

(A) 30 min

(B) 45 min

(C) 60 min

(D) 80 min

Ans. C**Sol.** $\frac{dy}{dt} = -k\sqrt{y}$; when $t = 0$; $y = 4$

$$\int_4^0 \frac{dy}{\sqrt{y}} = -k \int_0^t dt$$

$$2\sqrt{y} \Big|_4^0 = -kt = -\frac{t}{15}$$

$$0 - 4 = -\frac{t}{15} \Rightarrow t = 60 \text{ minutes} \Rightarrow (C)$$

3. The equation of the trajectories which is orthogonal to the family of curves $\cos y = ae^{-x}$ is :(A) $\sin y = ce^x$ (B) $\cos y = ce^x$ (C) $\sin y = ce^{-x}$

(D) None of these

चक्र $\cos y = ae^{-x}$ के लम्बवर्तीय पथ का समीकरण है :(A) $\sin y = ce^x$ (B) $\cos y = ce^x$ (C) $\sin y = ce^{-x}$

(D) कोई नहीं

Ans. C**Sol.** $+\sin y \frac{dy}{dx} = +ae^{-x}$

$$\sin y \frac{dy}{dx} = \cos y \Rightarrow \frac{dy}{dx} = \cot y$$

equation to the orthogonal trajectory will be obtained by solving the differential equation

$$-\frac{dx}{dy} = \cot y$$

$$c - x = \ln(\sin y) \Rightarrow \sin y = ce^{-x}$$

4. A function $y = f(x)$ satisfies the differential equation $\frac{dy}{dx} - y = \cos x - \sin x$, with initial condition that y is bounded when $x \rightarrow \infty$. The area enclosed by $y = f(x)$, $y = \cos x$ and the y -axis in the 1st quadrant :फलन $y = f(x)$, अवकल समीकरण $\frac{dy}{dx} - y = \cos x - \sin x$, को सन्तुष्ट करता है, तथा दिया है कि $x \rightarrow \infty$ पर परिबद्ध है, तो प्रथमप्रक्षेप $y = f(x)$, $y = \cos x$ तथा y -अक्ष द्वारा घिरा क्षेत्रफल होगा :

(A) $\sqrt{2} - 1$

(B) $\sqrt{2}$

(C) 1

(D) $\frac{1}{\sqrt{2}}$

Ans. A**Sol.** I.F. = e^{-x}

$$\therefore y e^{-x} = \int e^{-x} (\cos x - \sin x) dx \quad \text{put } -x = t$$

$$= - \int e^t (\cos t + \sin t) dt$$

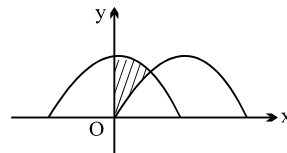
$$= - e^t \sin t + c$$

$$y e^{-x} = e^{-x} \sin x + c$$

since y is bounded when $x \rightarrow \infty \Rightarrow c = 0$

$$\therefore y = \sin x$$

$$\text{Area} = \int_0^{\pi/4} (\cos x - \sin x) dx = \sqrt{2} - 1 \Rightarrow \text{(A)}$$



5. A function $f(x)$ satisfying $\int_0^1 f(tx) dt = n f(x)$, where $x > 0$, is :

यदि फलन $f(x)$ इस प्रकार है कि $\int_0^1 f(tx) dt = n f(x)$ जहाँ $x > 0$ है, तो :

(A) $f(x) = c \cdot x^{\frac{1-n}{n}}$

(B) $f(x) = c \cdot x^{\frac{n}{n-1}}$

(C) $f(x) = c \cdot x^{\frac{1}{n}}$

(D) $f(x) = c \cdot x^{(1-n)}$

Ans. A

$$\text{Sol. } \int_0^1 f(tx) dt = n \cdot f(x)$$

$$\text{put } tx = y \Rightarrow dt = \frac{1}{x} dy$$

$$\therefore \frac{1}{x} \int_0^x f(y) dy = n f(x)$$

$$\therefore \int_0^x f(y) dy = x \cdot n \cdot f(x)$$

Differentiating

$$f(x) = n [f(x) + x f'(x)]$$

$$f(x) (1 - n) = n x f'(x)$$

$$\therefore \frac{f'(x)}{f(x)} = \frac{1-n}{n x}$$

$$\text{Integrating} \quad \ln f(x) = \left(\frac{1-n}{n} \right) \ln c x = \ln (c x)^{\frac{1-n}{n}}$$

$$\therefore f(x) = c x^{\frac{1-n}{n}}$$

6. A curve $y = f(x)$ passing through the point $\left(1, \frac{1}{\sqrt{e}}\right)$ satisfies the differential equation $\frac{dy}{dx} + x e^{-\frac{x^2}{2}} = 0$. Then which of the following does not hold good?
- (A) $f(x)$ is differentiable at $x = 0$
 (B) $f(x)$ is symmetric w.r.t. the origin
 (C) $f(x)$ is increasing for $x < 0$ and decreasing for $x > 0$
 (D) $f(x)$ has two inflection points

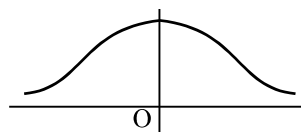
वक्र $y = f(x)$ बिन्दु $\left(1, \frac{1}{\sqrt{e}}\right)$ से गुजरता है, तथा अवकल समीकरण $\frac{dy}{dx} + x e^{-\frac{x^2}{2}} = 0$ को सन्तुष्ट करता है, तब निम्नलिखित में से

दिए गए विकल्पों में से

- (A) $x > 0$ पर फलन $f(x)$ अवकलनीय है
 (B) फलन $f(x)$ मूलबिन्दु के सापेक्ष सममित है
 (C) फलन $f(x)$, $x < 0$ के लिए वर्धमान तथा $x > 0$ के लिए ह्रासमान है
 (D) फलन $f(x)$ के दो नति परिवर्तन बिन्दु हैं

Ans. B

Sol. $y = f(x) = e^{-\frac{x^2}{2}}$. The graph is as shown



7. A curve passing through (2, 3) and satisfying the differential equation $\int_0^x t y(t) dt = x^2 y(x)$, ($x > 0$) is :

एक वक्र बिन्दु (2, 3) से गुजरता है तथा अवकल समीकरण $\int_0^x t y(t) dt = x^2 y(x)$, ($x > 0$), को सन्तुष्ट करता है :

- (A) $x^2 + y^2 = 13$ (B) $y^2 = \frac{9}{2} x$ (C) $\frac{x^2}{8} + \frac{y^2}{18} = 1$ (D) $xy = 6$

Ans. D

Sol. differentiate $x y(x) = x^2 y'(x) + 2xy(x)$
or $x y(x) + x^2 y'(x) = 0$

$$x \frac{dy}{dx} + y = 0$$

$$\ln y + \ln x = \ln c$$

$$xy = c \Rightarrow (D)$$

8. A function $y = f(x)$ satisfies $(x+1) \cdot f'(x) - 2(x^2+x)f(x) = \frac{e^{x^2}}{(x+1)}$, $\forall x > -1$. If $f(0) = 5$, then $f(x)$ is :

यदि फलन $y = f(x)$ सम्बन्ध $(x+1) \cdot f'(x) - 2(x^2+x)f(x) = \frac{e^{x^2}}{(x+1)}$, $\forall x > -1$ को सन्तुष्ट करता है तथा

$f(0) = 5$, तो $f(x)$ होगा :

(A) $\left(\frac{3x+5}{x+1}\right) \cdot e^{x^2}$ (B) $\left(\frac{6x+5}{x+1}\right) \cdot e^{x^2}$ (C) $\left(\frac{6x+5}{(x+1)^2}\right) \cdot e^{x^2}$ (D) $\left(\frac{5-6x}{x+1}\right) \cdot e^{x^2}$

Ans. B

Sol. $f'(x) - \frac{2x(x+1)}{x+1}f(x) = \frac{e^{x^2}}{(x+1)^2}$

$$I.F. = e^{\int -2x dx} = e^{-x^2}$$

$$\therefore f(x) \cdot e^{-x^2} = \int \frac{dx}{(x+1)^2} \Rightarrow f(x) \cdot e^{-x^2} = -\frac{1}{x+1} + C$$

$$\text{at } x=0, f(0)=5 \Rightarrow C=6$$

$$\therefore f(x) = \left(\frac{6x+5}{x+1}\right) \cdot e^{x^2}$$

9. The solution of primitive integral equation $(x^2+y^2)dy = xy dx$, is $y = y(x)$. If $y(1) = 1$ and $y(x_0) = e$, then x_0 is :

अवकल समीकरण $(x^2+y^2)dy = xy dx$ का हल $y = y(x)$, यदि $y(1) = 1$ एवं $y(x_0) = e$, को संतुष्ट करता हो, तो x_0 का मान है :

(A) $\sqrt{2(e^2-1)}$ (B) $\sqrt{2(e^2+1)}$ (C) $\sqrt{3} e$ (D) $\sqrt{\frac{e^2+1}{2}}$

Ans. C

Sol. $\frac{dy}{dx} = \frac{xy}{x^2 + y^2}$ This is homogeneous so put $y = vx$

$$\frac{dy}{dx} = v + \frac{dv}{dx}x$$

$$v + \frac{dv}{dx}x = \frac{vx \cdot vx}{x^2 + (vx)^2} = \frac{v}{1 + v^2}$$

$$\text{or } x \frac{dv}{dx} = \frac{v}{1 + v^2} - \frac{v}{1} = \frac{-v^3}{1 + v^2}$$

$$\int -\frac{(1 + v^2)dv}{v^3} = \int \frac{dx}{x} \quad \text{or} \quad -\int \frac{dv}{v^3} - \int \frac{dv}{v} = \int \frac{dx}{x}$$

$$\text{or } \frac{1}{2v^2} - \log v = \log x + c$$

$$\text{put } v = \frac{y}{x} \Rightarrow \frac{x^2}{2y^2} = \log \frac{y}{x} + \log x + c = \log y + c \text{ given } y(1) = 1$$

$$\frac{1}{2} = c \Rightarrow \frac{x^2}{2y^2} = \log y + \frac{1}{2}$$

Now put $x = x_0$, $y = e$

$$\frac{x_0^2}{2e^2} = 1 + \frac{1}{2} = \frac{3}{2} \quad \text{or} \quad x_0^2 = \frac{6e^2}{2} = 3e^2$$

$$\Rightarrow x_0 = \sqrt{3} \cdot e$$

10. For the primitive integral equation $ydx + y^2dy = xdy$; $x \in \mathbb{R}$, $y > 0$, $y = y(x)$, $y(1) = 1$, then $y(-3)$ is :

आद्य समाकल समीकरण $ydx + y^2dy = xdy$; $x \in \mathbb{R}$, $y > 0$, $y = y(x)$, $y(1) = 1$ के लिये $y(-3)$ होगा :

- (A) 3 (B) 2 (C) 1 (D) 5

Ans. A

Sol. $\frac{xdy - ydx}{y^2} = dy$ or $\frac{ydx - xdy}{y^2} = -dy$

$$\text{or } \int d\left(\frac{x}{y}\right) = -\int dy$$

$$\frac{x}{y} = -y + c \text{ Given } y(1) = 1 \Rightarrow 1 = -1 + c \Rightarrow c = 2$$

$$\frac{x}{y} = -y + 2 \text{ Now } y(-3), \frac{-3}{y} = -y + 2$$

$$\text{or } y^2 - 2y - 3 = 0$$

$$y^2 - 3y + y - 3 = 0 = (y - 3)(y + 1) = 0$$

$$y = 3 \text{ or } y = -1 \text{ But } y > 0$$

$$\therefore y = 3$$

11. The solution of the differential equation $2x^2y \frac{dy}{dx} = \tan(x^2y^2) - 2xy^2$ given $y(1) = \sqrt{\frac{\pi}{2}}$ is :

अवकल समीकरण $2x^2y \frac{dy}{dx} = \tan(x^2y^2) - 2xy^2$ का हल होगा, दिया गया है $y(1) = \sqrt{\frac{\pi}{2}}$:

(A) $\sin(x^2y^2) - 1 = 0$

(B) $\cos\left(\frac{\pi}{2} + x^2y^2\right) + x = 0$

(C) $\sin(x^2y^2) = e^{x-1}$

(D) $\sin(x^2y^2) = e^{2(x-1)}$

Ans. C

Sol. $2x^2y \frac{dy}{dx} = \tan(x^2y^2) - 2xy^2$

$$2x^2y \frac{dy}{dx} + 2xy^2 = \tan(x^2y^2)$$

$$x^2y^2 = t$$

$$2xy^2 + 2x^2y \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} = \tan t$$

$$\int \cot t dt = \int \frac{1}{x} dx$$

$$\ln |\sin t| = x + c$$

$$\ln(\sin(x^2y^2)) = x + c$$

$$\left(1, \sqrt{\frac{\pi}{2}}\right)$$

$$\ln \left[\sin \frac{\pi}{2} \right] = 1 + c$$

$$0 = 1 + c \Rightarrow c = -1$$

$$\sin(x^2 y^2) = (e)^{x-1}$$

12. The general solution of the differential equation $\frac{dy}{dx} + x(x+y) = x(x+y)^3 - 1$ is : (where C is arbitrary constant.)

अवकल समीकरण $\frac{dy}{dx} + x(x+y) = x(x+y)^3 - 1$ का व्यापक हल होगा : (जहाँ C एक स्वेच्छ अचर है।)

$$(A) \ln \left| \frac{(x+y+1)(x+y-1)}{(x+y)^4} \right| = x^2 + C$$

$$(B) \ln \left| \frac{(x+y+1)(x+y-1)}{(x+y)^2} \right| = x^2 + C$$

$$(C) 2 \ln \left| \frac{(x+y+1)(x+y-1)}{(x+y)^2} \right| = x^2 + C$$

$$(D) \ln \left| \frac{(x+y+1)(x+y-1)}{(x+y)^2} \right| = x + C$$

Ans. B

Sol. $\frac{dy}{dx} + x(x+y) = x(x+y)^3 - 1$

$$1 + \frac{dy}{dx} + x(x+y) = x(x+y)^3$$

$$x + y = t$$

$$1 + \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} + xt = xt^3$$

$$\frac{dt}{dx} = x(t^3 - t)$$

$$\frac{dt}{t(t^2 - 1)} = x dx$$

$$\frac{dt}{t(t-1)(t+1)} = \frac{A}{t} + \frac{B}{t-1} + \frac{C}{t+1}$$

$$1 = A(t-1)(t+1) + B(t)(t+1) + C(t)(t-1)$$

$$t = 0$$

$$t = 1$$

$$t = -1$$

$$A = -1$$

$$B = \frac{1}{2}$$

$$C = \frac{1}{2}$$

$$\int \left(-\frac{dt}{t} + \frac{1}{2} \frac{dt}{t-1} + \frac{1}{2} \frac{dt}{t+1} \right) = \int x dx$$

$$\int \frac{-2dt}{t} + \int \frac{dt}{t-1} + \int \frac{dt}{t+1} = \int 2x dx$$

$$-2 \ln |t| + \ln |t-1| + \ln |t+1| = x^2 + c$$

$$\ln \left(\frac{(t-1)(t+1)}{t^2} \right) = x^2 + c$$

$$\ln \left[\frac{(x+y-1)(x+y+1)}{(x+y)^2} \right] = x^2 + c$$

13. The solution of the equation $\frac{dy}{dx} + \frac{1}{x} \tan y = \frac{1}{x^2} \tan y \sin y$ is :

समीकरण $\frac{dy}{dx} + \frac{1}{x} \tan y = \frac{1}{x^2} \tan y \sin y$ का हल होगा :

(A) $2y = \sin y (1 - 2cx^2)$

(B) $2x = \cot y (1 + 2cx^2)$

(C) $2x = \sin y (1 + 2cx^2)$

(D) $2x \sin y = 1 - 2cx^2$

Ans. C

Sol. $\frac{dy}{dx} + \frac{1}{x} \tan y = \frac{1}{x^2} \tan y \sin y$

$$\cot y \operatorname{cosec} y \frac{dy}{dx} + \frac{1}{x} \operatorname{cosec} y = \frac{1}{x^2}$$

$$\operatorname{cosec} y = t$$

$$\operatorname{cosec} y \cot y \frac{dy}{dx} = \frac{-dt}{dx}$$

$$\frac{-dt}{dx} + \frac{t}{x} = \frac{1}{x^2}$$

$$\frac{dt}{dx} + \left(\frac{-1}{x} \right) t = \frac{-1}{x^2}$$

$$\text{I.F.} = e^{\int \frac{-1}{x} dx} = e^{-\ln x} = \frac{1}{x}$$

$$\int \frac{1}{x} \frac{dt}{dx} - \frac{1}{x^2} t = \int -\frac{1}{x^3}$$

$$\frac{t}{x} = -\frac{x^{-3+1}}{-3+1} + C$$

$$\frac{\operatorname{cosec} y}{x} = \frac{1}{2x^2} + C$$

$$2x \operatorname{cosec} y = 1 + 2Cx^2$$

$$2x = (1 + 2Cx^2) \sin y$$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

14. The equation of the curve such that the distance between the origin and the tangent line at an arbitrary point is equal to the distance between the origin and the normal at the same point is : (where c is an arbitrary constant)

उस वक्र का समीकरण, जिस पर खींची गई स्पर्श रेखा पर स्थित किसी बिन्दु की मूल बिन्दु से दूरी, उसी बिन्दु पर खींचे अभिलम्ब पर मूल बिन्दु से दूरी के बराबर हो (जहाँ c प्राचल है)

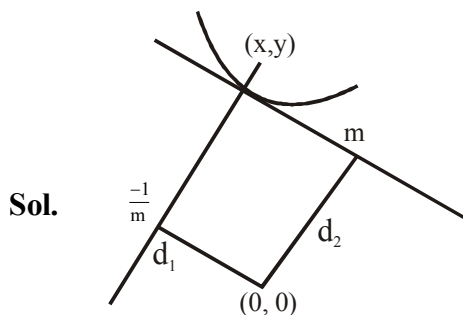
(A) $x^2 + y^2 = c \cdot e^{\arctan(y/x)}$

(B) $\sqrt{x^2 + y^2} = c \cdot e^{-\arctan(y/x)}$

(C) $\sqrt{x^2 + y^2} = c \cdot e^{\arctan(y/x)}$

(D) $x^2 + y^2 = c \cdot e^{-\arctan(y/x)}$

Ans. BC



$$Y - y = -\frac{1}{m}(X - x)$$

$$X + mY - my - x = 0$$

$$d_1 = d_2$$

$$(0, 0)$$

$$d_1 = \left| \frac{x + my}{\sqrt{m^2 + 1}} \right|$$

$$Y - y = m(X - x)$$

$$mX - Y - mx + y = 0$$

$$(0, 0)$$

$$d_2 = \left| \frac{y - mx}{\sqrt{m^2 + 1}} \right|$$

$$\frac{|x + my|}{\sqrt{m^2 + 1}} = \frac{|y - mx|}{\sqrt{m^2 + 1}}$$

$$x + my = \pm (y - mx)$$

$$x + my = y - mx$$

$$m(y + x) = y - x$$

$$\frac{dy}{dx} = \frac{y - x}{x + y}$$

$$y = vx$$

$$v + \frac{xdv}{dx} = \frac{v - 1}{v + 1}$$

$$\frac{xdv}{dx} = \frac{v - 1 - v^2 - v}{v + 1}$$

$$\frac{xdv}{dx} = -\frac{(v^2 + 1)}{v + 1}$$

$$\frac{v + 1}{v^2 + 1} dv = -\frac{dx}{x}$$

$$\int \frac{v dv}{v^2 + 1} + \int \frac{dv}{v^2 + 1} = -\ln x + c$$

$$\frac{1}{2} \ln(v^2 + 1) + \tan^{-1} v = -\ln x + C$$

$$x + my = mx - y$$

$$m(x - y) = x + y$$

$$\frac{dy}{dx} = \frac{x + y}{x - y}$$

$$v + \frac{xdv}{dx} = \frac{v + 1}{1 - v}$$

$$\frac{xdv}{dx} = \frac{v + 1}{1 - v} - v = \frac{v + 1 - v + v^2}{1 - v} = \frac{v^2 + 1}{1 - v}$$

$$\int \frac{1 - v}{v^2 + 1} dv = \int \frac{dx}{x}$$

$$\int \frac{dv}{v^2 + 1} - \frac{1}{2} \int \frac{2v}{v^2 + 1} dv = \ln x + C$$

$$\tan^{-1} v - \ln \sqrt{v^2 + 1} = \ln x + C$$

$$\tan^{-1} \left(\frac{y}{x} \right) + c = \ln \sqrt{y^2 + x^2}$$

$$\sqrt{x^2 + y^2} = ce^{\tan^{-1}\left(\frac{y}{x}\right)}$$

$$\ln \sqrt{\frac{y^2 + x^2}{x^2}} + \ln x = -\tan^{-1} \frac{y}{x} + c$$

$$\ln(\sqrt{x^2 + y^2}) + \ln C = -\tan^{-1}\left(\frac{y}{x}\right)$$

$$\sqrt{x^2 + y^2} = ce^{-\tan^{-1}\left(\frac{y}{x}\right)}$$

15. Let $\frac{dy}{dx} + y = f(x)$ where y is a continuous function of x with $y(0) = 1$ and $f(x) = \begin{cases} e^{-x} & \text{if } 0 \leq x \leq 2 \\ e^{-2} & \text{if } x > 2 \end{cases}$. Which of the following hold(s) good :

यदि $\frac{dy}{dx} + y = f(x)$ जहाँ y, x में सतत फलन है, $y(0) = 1$ तथा $f(x) = \begin{cases} e^{-x} & \text{यदि } 0 \leq x \leq 2 \\ e^{-2} & \text{यदि } x > 2 \end{cases}$ तब निम्नलिखित में कौनसे

दिए गए सही हैं

- (A) $y(1) = 2e^{-1}$ (B) $y'(1) = -e^{-1}$ (C) $y(3) = -2e^{-3}$ (D) $y'(3) = -2e^{-3}$

Ans. ABD

Sol. $\frac{dy}{dx} + y = f(x)$

I.F. = e^x

$$ye^x = \int e^x f(x) dx + C$$

$$\text{now if } 0 \leq x \leq 2 \text{ then } ye^x = \int e^x e^{-x} dx + C \Rightarrow ye^x = x + C$$

$$x = 0, y(0) = 1, \quad C = 1$$

$$\therefore ye^x = x + 1 \quad \dots(1)$$

$$y = \frac{x+1}{e^x}; \quad y(1) = \frac{2}{e} \text{ Ans. } \Rightarrow \quad (A) \text{ is correct}$$

$$y' = \frac{e^x - (x+1)e^x}{e^{2x}}$$

$$y'(1) = \frac{e - 2e}{e^2} = \frac{-e}{e^2} = -\frac{1}{e} \text{ Ans. } \Rightarrow \quad (B) \text{ is correct}$$

if $x > 2$

$$ye^x = \int e^{x-2} dx$$

$$ye^x = e^{x-2} + C$$

$$y = e^{-2} + Ce^{-x}$$

as y is continuous

$$\therefore \lim_{x \rightarrow 2} \frac{x+1}{e^x} = \lim_{x \rightarrow 2} (e^{-2} + Ce^{-x})$$

$$3e^{-2} = e^{-2} + Ce^{-2} \Rightarrow C = 2$$

$$\therefore \text{for } x > 2$$

$$y = e^{-2} + 2e^{-x} \text{ hence } y(3) = 2e^{-3} + e^{-2} = e^{-2}(2e^{-1} + 1)$$

$$y' = -2e^{-x}$$

$$y'(3) = -2e^{-3} \text{ Ans. } \Rightarrow \text{(D) is correct}$$

16. If $y(x)$ satisfies the differential equation $\frac{dy}{dx} = \sin 2x + 3y \cot x$ and $y\left(\frac{\pi}{2}\right) = 2$ then which of the following statement(s) is/are correct?

$$(A) y\left(\frac{\pi}{6}\right) = 0$$

$$(B) y'\left(\frac{\pi}{3}\right) = \frac{9-3\sqrt{2}}{2}$$

$$(C) y(x) \text{ increases in the interval } \left(0, \frac{\pi}{2}\right)$$

$$(D) \int_{-\pi/2}^{\pi/2} y(x) dx = x$$

यदि $y(x)$ अवकल समीकरण $\frac{dy}{dx} = \sin 2x + 3y \cot x$ तथा $y\left(\frac{\pi}{2}\right) = 2$ को संतुष्ट करता है, तो निम्न में से कौनसा/कौनसे कथन

1 R g l x 8

$$(A) y\left(\frac{\pi}{6}\right) = 0$$

$$(B) y'\left(\frac{\pi}{3}\right) = \frac{9-3\sqrt{2}}{2}$$

$$(C) y(x) \text{ अन्तराल } \left(0, \frac{\pi}{2}\right) \text{ में वर्धमान होगा}$$

$$(D) \int_{-\pi/2}^{\pi/2} y(x) dx = x$$

Ans. A

Sol. $\frac{dy}{dx} = \sin 2x + 3y \cot x$

$$\frac{dy}{dx} + y(-3 \cot x) = \sin 2x$$

$$\text{I.E.} = e^{\int -3 \cot x dx} = e^{-3 \ln \sin x}$$

$$= \frac{1}{\sin^3 x}$$

$$\int \frac{1}{\sin^3 x} \frac{dy}{dx} + y \left(\frac{-3 \cot x}{\sin^3 x} \right) = \int \frac{2 \sin x \cos x}{\sin^2 x}$$

$$\frac{y}{\sin^3 x} = 2 \int \frac{dx}{t^2}; \quad t = \sin x$$

$$\frac{y}{\sin^3 x} = \frac{-2}{\sin x} + C \quad \left(\frac{\pi}{2}, 2 \right)$$

$$C = +4 \quad y = -2 \sin^2 x + 4 \sin^3 x$$

$$f\left(\frac{\pi}{6}\right) = -2 \times \frac{1}{4} + 4 \times \frac{1}{8}$$

$$= -\frac{1}{2} + \frac{1}{2} = 0$$

$$f'(x) = -4 \sin x \cos x + 12 \sin^2 \cos x = 4 \sin x \cos x (3 \sin x - 1)$$

$$\Rightarrow -4 + \frac{\sqrt{3}}{2} \times \frac{1}{2} + 12 \times \frac{3}{4} \times \frac{1}{2}$$

$$\Rightarrow -\sqrt{3} + \frac{9}{2}$$

$$\Rightarrow \frac{9 - 2\sqrt{3}}{2}$$

17. Solve : $\tan y \frac{dy}{dx} + \tan x = \cos y \cdot \cos^3 x$:

$\tan y \frac{dy}{dx} + \tan x = \cos y \cdot \cos^3 x$ को हल होगा :

$$(A) \sec x \sec y = \frac{1}{2} \left(x + \frac{1}{2} \sin 2x \right) + C$$

$$(B) ((x + \sin x \cos x) + 2C) \cos x \cos y = 2$$

$$(C) \cos x \cos y = \frac{1}{2} (x + \sin x \cos x) + C$$

$$(D) \sec x \sec y = \frac{1}{2} (x + \cos x) + C$$

Ans. AB

Sol. $\tan y \frac{dy}{dx} + \tan x = \cos y \cdot \cos^3 x$

$$\frac{\sin y}{\cos y} \frac{dy}{dx} + \frac{\sin x}{\cos x} = \cos y \cos^3 x$$

$$\cos x \sin y \frac{dy}{dx} + \sin x \cos y = \cos^2 y \cos^4 x$$

$$\div \cos^2 x \cos^2 y$$

$$\sec x \tan y \sec y \frac{dy}{dx} + \sec y \tan x \sec x = \cos^2 x = \frac{1 + \cos 2x}{2}$$

$$\sec x \sec y = \int \frac{1 + \cos 2x}{2} dx$$

$$\sec x \sec y = \frac{x}{2} + \frac{\sin 2x}{4} + C$$

$$\frac{1}{\cos x \cos y} = \frac{x}{2} + \frac{\sin x \cos x}{2} + C$$

$$2 = (x + \sin x \cos x + 2c) \cos x \cos y$$

18. The solution(s) of the differential equation $9(2-y)^2 \left(\frac{dy}{dx} \right)^2 = 4(3-y)$ is/are :

अवकल समीकरण $9(2-y)^2 \left(\frac{dy}{dx} \right)^2 = 4(3-y)$ का हल होगा :

(A) $(3-x)^2 = y^2(y+c)$

(B) $(x+c)^2 = y^2(3-y)$

(C) $y = 0$

(D) $y = 3$

Ans. BD

Sol. $9(2-y)^2 \left(\frac{dy}{dx} \right)^2 = 4(3-y)$

$$3(2-y) \frac{dy}{dx} = \pm 2(\sqrt{3-y})$$

$$+ \Rightarrow 3(2-y) \frac{dy}{dx} = 2\sqrt{3-y}$$

$$\frac{1+2-y-1}{\sqrt{3-y}} dy = \frac{2}{3} dx$$

$$\frac{(3-y)-1}{\sqrt{3-y}} dy = \frac{2}{3} dx$$

$$\int \left(\sqrt{3-y} - \frac{1}{\sqrt{3-y}} \right) dy = \int \frac{2}{3} dx$$

$$\frac{(3-y)^{3/2}}{-\frac{3}{2}} - \frac{\sqrt{3-y}}{-\frac{1}{2}} = \frac{2x}{3} + C$$

$$\Rightarrow 2\sqrt{3-y} - \frac{2}{3}(3-y)^{3/2} = \frac{2x}{3} + C$$

$$\Rightarrow 3\sqrt{3-y} - (3-y)^{3/2} = x + c$$

$$(\sqrt{3-y})(3-3+y) = x + c$$

$$y\sqrt{3-y} = x + c$$

$$y^2(3-y) = (x+c)^2$$

$$- \Rightarrow 3(2-y) \frac{dy}{dx} = -2\sqrt{3-y}$$

$$\frac{1+2-y-1}{\sqrt{3-y}} dy = -\frac{2}{3} dx$$

$$\int \left(\sqrt{3-y} - \frac{1}{\sqrt{3-y}} \right) dy = \int -\frac{2}{3} dx$$

$$\frac{(3-y)^{3/2}}{-\frac{3}{2}} - \frac{\sqrt{3-y}}{-\frac{1}{2}} = -\frac{2x}{3} + C$$

$$\frac{2(3-y)^{3/2}}{3} - 2\sqrt{3-y} = \frac{2x}{3} + C$$

$$(3-y)^{3/2} - 3\sqrt{3-y} = x + c$$

$$(\sqrt{3-y})[3-y-3]x + c$$

$$-y\sqrt{3-y} = x + c$$

19. Let $y = f(x)$ be a real valued function satisfying $x \frac{dy}{dx} = x^2 + y - 2$, $f(1) = 1$, then :

(A) $f(x)$ is minimum at $x = 1$

(B) $f(x)$ is maximum at $x = 1$

(C) $f(3) = 5$

(D) $f(2) = 3$

माना $y = f(x)$ एक वास्तविक मान फलन जो $x \frac{dy}{dx} = x^2 + y - 2$, $f(1) = 1$, को सन्तुष्ट करता है तो :

(A) $x > 1$ पर $f(x)$ न्यूनतम है

(B) $x > 1$ पर $f(x)$ अधिकतम है

(C) $f(3) = 5$

(D) $f(2) = 3$

Ans. AC

Sol. $x \frac{dy}{dx} = x^2 + y - 2$

$$\frac{dy}{dx} = x + \frac{y}{x} - \frac{2}{x}$$

$$\frac{dy}{dx} + \left(-\frac{1}{x}\right)y = x - \frac{2}{x}$$

$$I. F. = e^{\int -\frac{1}{x} dx} = e^{-\ln x} = \frac{1}{x}$$

$$\int \frac{1}{x} \frac{dy}{dx} + \left(-\frac{1}{x^2}\right)y = \int 1 - \frac{2}{x^2}$$

$$\frac{y}{x} = x + \frac{2}{x} + C$$

$$(1,1)$$

$$1 = 1 + 2 + C$$

$$C = -2$$

$$y = x^2 + 2 - 2x$$

$$y = x^2 - 2x + 2$$

$$x = 3$$

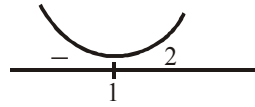
$$x = 2$$

$$y = 9 - 6 + 2$$

$$= 5$$

$$y' = 2x - 2$$

$$= 2(x - 1)$$



$$y = 4 - 4 + 2$$

$$y = 2$$

20. Solution of differential equation $x \cos x \left(\frac{dy}{dx} \right) + y(x \sin x + \cos x) = 1$ is : (where c is arbitrary constant)

- (A) $xy = \sin x + c \cos x$ (B) $xy \sec x = \tan x + c$
 (C) $xy + \sin x + c \cos x = 0$ (D) None of these

अवकल समीकरण $x \cos x \left(\frac{dy}{dx} \right) + y(x \sin x + \cos x) = 1$ का हल होगा : (जहाँ c एक स्वच्छ अचर है)

- (A) $xy = \sin x + c \cos x$ (B) $xy \sec x = \tan x + c$
 (C) $xy + \sin x + c \cos x = 0$ (D) इनमें से कोई नहीं

Ans. AB

Sol. $x \cos x \left(\frac{dy}{dx} \right) + y(x \sin x + \cos x) = 1$

$$\frac{dy}{dx} + y \left(\frac{2 \sin x + \cos x}{x \cos x} \right) = \frac{1}{x \cos x}$$

$$I. f = e^{\int \frac{x \sin x + \cos x}{x \cos x} dx}$$

$$= e^{\int \tan x + \frac{1}{x}} = e^{\ln \sec x + \ln x}$$

$$= x \sec x$$

$$\int x \sec x \frac{dy}{dx} + (x \sec x) y \left(\frac{2 \sin x + \cos x}{2 \cos x} \right) = \int \frac{x \sec x}{x \cos x} dx$$

$$yx \sec x = \tan x + c$$

$$xy = \sin x + c \cos x$$

21. If $y = y(x)$ is solution of differential equation $(1 - x^2) \frac{dx}{dy} + xy = x$ such that $y(1) = 1$ and $g(x) = (y - 1)^2$, then :

(A) $g(x)$ is an even function

$$(B) \lim_{x \rightarrow 1} \frac{g(x)}{(x-1)^2} = 2$$

(C) $g(x)$ has only two local minima

(D) $g(x) = 1$ has four solution

यदि $y = y(x)$ अवकल समीकरण $(1-x^2) \frac{dx}{dy} + xy = x$ का हल इस प्रकार है कि $y(1) = 1$ तथा $g(x) = (y-1)^2$ हो, तब :

(A) ह; गद्धसम फलन है

$$(B) \lim_{x \rightarrow 1} \frac{g(x)}{(x-1)^2} = 2$$

(C) ह; गद्धके केवल दो स्थानीय निम्निष्ठ है

(D) ह; गद्ध 1 के चार हल है

Ans. ABCD

Sol. $(1-x^2) \frac{dx}{dy} + xy = x$

$$(1-x^2) \frac{dx}{dy} = x(1-y)$$

$$\left(\frac{1-x^2}{x} \right) dx = (1-y) dy$$

$$y - \frac{y^2}{2} = \ln x - \frac{x^2}{2} + C$$

$$(1, 1)$$

$$1 - \frac{1}{2} = 0 - \frac{1}{2} + C$$

$$c = 1$$

$$y - \frac{y^2}{2} = \ln x - \frac{x^2}{2} + 1$$

$$2y - y^2 = 2\ln x - x^2 + 2$$

$$y^2 - 2y = x^2 - \ln x^2 - 2$$

$$(y-1)^2 = x^2 - \ln x^2 - 1$$

$$g(x) = x^2 - \ln x^2 - 1$$

Even

$$\lim_{x \rightarrow 1} \frac{x^2 - \ln x^2 - 1}{(x-1)^2} = \frac{2x - \frac{1}{x^2} \times 2x}{2(x-1)}$$

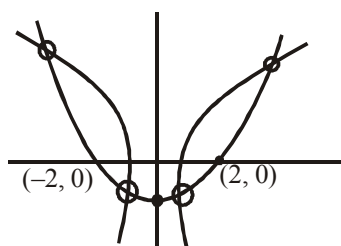
$$\Rightarrow \frac{2x \left(1 - \frac{1}{x^2}\right)}{2(x-1)}$$

$$\Rightarrow \frac{x(x^2 - 1)}{(x-1)x^2} = \frac{(x-1)(x+1)}{(x-1)x}$$

$$\lim_{x \rightarrow 1} \frac{x+1}{x} = 2$$

$$x^2 - \ln x^2 - 1 = 1 \qquad g'(x) = 2x - \frac{2x}{x^2}$$

$$x^2 = 2(\ln x + 2) \qquad = 2 \left(\frac{x^2 - 1}{x} \right) = \frac{(x-1)(x+1)}{x}$$



$$x^2 = 2(\ln x + 2)$$

$$y = \frac{x^2}{2} - 2 = \ln|x|$$